SMART GAIT USING AI:
Human Identification using Deep Learning

Internship Report

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by

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Reference Note

This internship report titled **"Smart Gait Using AI"** contains information, figures, and concepts referenced from authentic academic and research sources. All referenced materials are properly cited in the **Bibliography** section with their corresponding reference numbers. Wherever external content, figures, or ideas have been adapted or reproduced, appropriate credit and reference numbers have been included to ensure academic integrity and to avoid copyright violations.

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Abstract

Human identification through gait analysis offers a unique, non-intrusive, and distance-based biometric approach that remains effective even when conventional modalities such as face or fingerprint recognition fail. This project, titled **"Smart Gait Using AI: Human Identification using Deep Learning"**, focuses on developing a complete end-to-end framework for gait-based recognition using artificial intelligence and computer vision techniques.

The proposed system employs the YOLOv8 model for accurate real-time human detection and segmentation from video frames, followed by silhouette extraction using background subtraction and morphological filtering. A sequence of extracted silhouettes is then converted into a **Gait Energy Image (GEI)**, which compactly represents both spatial and temporal walking characteristics of each subject. These GEIs are fed into a customized Convolutional Neural Network (CNN) designed to learn discriminative gait patterns for identity classification.

The system was evaluated using the **CASIA-B** dataset under three conditions — normal walking, carrying a bag, and wearing a coat — achieving an overall recognition accuracy of **97.5%**, outperforming several baseline approaches such as GEINet, GaitSet, and GaitPart. The model demonstrated robustness against moderate viewpoint and clothing variations, validating the potential of gait as a reliable biometric identifier.

The Smart Gait AI framework provides a scalable foundation for applications in **security, surveillance, healthcare, and forensic analysis**. It bridges the gap between traditional handcrafted gait analysis and modern deep-learning-based recognition systems, enabling future research toward real-time, multi-view, and privacy-preserving gait recognition technologies.

1 Introduction

1.1 Background

Biometric recognition refers to the automatic identification of individuals based on unique physiological or behavioral characteristics. Conventional biometric methods such as fingerprint, facial, or iris recognition have achieved high accuracy in controlled environments, yet they often require close-range cooperation and good illumination. These limitations restrict their application in surveillance and remote monitoring scenarios where subjects may be far from the camera or partially occluded.

Gait, defined as the manner of walking, provides an alternative biometric cue that can be captured unobtrusively at a distance. Every individual possesses a distinctive walking pattern influenced by body shape, posture, limb coordination, and rhythm. Because gait does not require subject cooperation and can be observed from afar, it has emerged as an effective modality for identifying people in outdoor or security contexts. Unlike fingerprints or facial images, gait features are difficult to disguise intentionally, making gait recognition a powerful complement to existing biometric systems.

1.2 Definition of Gait Recognition

Gait recognition is the process of identifying or verifying a person's identity by analyzing their walking pattern from video sequences. It involves capturing motion data, extracting spatio–temporal features that describe body movement, and comparing these features against a stored database. Over the last two decades, gait recognition has progressed from handcrafted model-based methods to appearance-based approaches and, more recently, to deep learning models capable of automatically learning discriminative features.

Model-based approaches reconstruct the human body as an articulated structure of joints and limbs, estimating motion parameters such as stride length, hip angle, or velocity. These methods can achieve view invariance but require accurate pose estimation and high-resolution imagery. Appearance-based approaches, on the other hand, operate directly on binary silhouettes or spatio-temporal templates such as the Gait Energy Image (GEI). They are simpler to compute and robust to low-quality videos, making them suitable for large-scale surveillance datasets.

1.3 Need for Gait Recognition

With the widespread installation of CCTV networks and the increasing need for contactless identification, gait recognition provides a non-invasive mechanism to enhance public safety and monitoring. In forensic applications, suspects can be recognized even when faces are hidden or when gloves obscure fingerprints. In healthcare, gait analysis supports diagnosis of neurological disorders such as Parkinson's disease or post-stroke mobility assessment. The same technology can also be applied to human–computer interaction, where a system adapts to a user's identity or physical state by observing walking behavior.

Despite these advantages, gait recognition still faces challenges due to viewpoint changes, clothing variations, and background noise. Addressing these factors requires advanced preprocessing, silhouette normalization, and deep learning models capable of extracting invariant representations.

1.4 Motivation for the Project

Recent advances in computer vision, particularly convolutional neural networks (CNNs) and real-time object detectors such as YOLO, have transformed how visual data can be analyzed. These methods automatically learn complex spatial features, eliminating the need for manual feature design. Integrating such techniques with gait recognition offers a pathway to develop a smart, AI-driven system that can operate automatically and accurately in practical environments.

Traditional handcrafted pipelines relied heavily on manual silhouette extraction and feature computation, which limited robustness. By contrast, an end-to-end deep learning model can handle noise, illumination changes, and subtle variations more effectively. The goal of this project is to design a complete pipeline—called **Smart Gait AI**—that combines real-time detection, silhouette processing, and deep learning-based person identification into a single framework.

1.5 Overview of Smart Gait AI

The Smart Gait AI system consists of several key stages:

1. **Human Detection:** Using the YOLOv8 object detection model to locate individuals in each frame of a video sequence.

2. **Silhouette Extraction:** Applying background subtraction, thresholding, and morphological filtering to obtain clean binary silhouettes.

3. **Feature Generation:** Converting sequences of silhouettes into a single Gait Energy Image (GEI) that represents the average motion pattern of a walking cycle.

4. **Deep Feature Learning:** Feeding GEI images into a CNN that automatically learns discriminative spatial features for each individual.

5. **Classification:** The trained CNN outputs the predicted identity, which can then be used for authentication or tracking.

A schematic of the pipeline is illustrated below.

1.6 Applications of Gait Recognition

Gait recognition has diverse applications across domains:

- **Security and Surveillance:** Continuous monitoring of public areas, airports, and campuses to detect or identify individuals of interest.
- **Forensic Analysis:** Identification of criminals or missing persons based on body movement captured in CCTV footage.
- **Healthcare and Rehabilitation:** Monitoring patient mobility, postural stability, and recovery progress in clinical settings.
- **Smart Access Control:** Implementing hands-free, contactless access systems based on a person's walk pattern.
- **Human–Robot Interaction:** Enabling robots or smart devices to personalize responses based on user recognition and physical state.

1.7 Objectives of the Project

The primary objectives of this project are:

- To develop an automated system for gait-based human identification using AI.
- To implement silhouette extraction and preprocessing using YOLOv8 and morphological operations.
- To generate Gait Energy Images (GEIs) as compact spatio-temporal descriptors.
- To design and train a CNN model capable of high recognition accuracy.

- To evaluate the system using the CASIA-B dataset under multiple conditions.

1.8 Scope of the Work

The scope of Smart Gait AI focuses on two-dimensional appearance-based recognition using side-view walking sequences. The system is designed primarily for controlled video datasets, though the techniques can be extended to real-world footage. Future work may incorporate 3-D pose estimation, temporal modeling with LSTMs or transformers, and multi-view integration.

1.9 Advantages of the Proposed Approach

- Non-intrusive and contactless identification at a distance.
- Works when facial or fingerprint features are unavailable.
- Resistant to partial occlusion and illumination variation.
- Scalable to large populations using efficient CNN inference.
- Can be integrated into existing CCTV infrastructure.

1.10 Limitations and Assumptions

The current system assumes a relatively static background and consistent walking direction. Viewpoint changes greater than 90° and drastic clothing differences can still affect accuracy. These factors are addressed partly by normalization and data augmentation but remain open research challenges.

1.11 Organization of Remaining Chapters

The rest of this report is organized as follows. Chapter 2 states the detailed problem formulation and challenges. Chapter 3 explains the process of silhouette extraction and preprocessing. Chapter 4 reviews various feature extraction techniques and datasets used in gait recognition. Chapter 5 focuses on the exploitation of silhouette and edge characteristics. Chapter 6 presents the deep learning model, training, and evaluation methodology. Chapter 7 discusses experimental results and comparisons. Chapter 8 concludes the report and outlines future research directions.

2 Problem Statement

2.1 Introduction

Human identification is a critical requirement in domains such as security, surveillance, access control, and healthcare. Traditional biometric systems—fingerprint, face, and iris recognition—depend heavily on cooperation, high-quality images, and controlled environments. In unconstrained situations, such as low-light surveillance or when the face is partially obscured, these modalities often fail to provide reliable results.

Gait recognition offers a unique alternative. It relies on the analysis of human walking patterns, which are difficult to conceal or imitate deliberately. Each individual's gait results from a combination of skeletal structure, muscle coordination, and posture, creating a distinctive signature. However, designing a system that extracts this signature accurately from video sequences remains challenging because of variations in background, clothing, lighting, and view angle.

This chapter defines the problem addressed by the Smart Gait AI project, identifies the challenges involved, and describes the objectives, scope, and expected outcomes.

2.2 Problem Definition

The goal of this project is to design and implement an automated system that can accurately identify individuals based solely on their walking pattern, even under varying environmental and recording conditions. Formally, given a video sequence $V = \{f_1, f_2, \ldots, f_T\}$ of a person walking, the system must compute a compact and discriminative gait representation $G(V)$ and determine the subject's identity I by learning a mapping function:

$$I = \mathcal{M}(G(V); \theta) - (1)$$

where $\mathcal{M}$ is a deep learning model parameterized by θ. The system should be robust to common variations such as:

- Clothing and carrying conditions (e.g., coat, bag).
- Viewpoint changes (frontal, side, oblique angles).
- Illumination, background clutter, and camera noise.

2.3 Existing System Limitations

Existing gait recognition methods often suffer from one or more of the following limitations:

1. **Manual preprocessing:** Earlier approaches depend on background subtraction tuned manually for each scene, reducing scalability.
2. **View and appearance dependency:** Hand-crafted features such as contour length or stride width vary drastically with camera angle.
3. **Shallow feature extraction:** Traditional classifiers like SVM or k-NN cannot capture complex spatio-temporal variations in walking style.
4. **Limited dataset generalization:** Many models trained on small, controlled datasets fail on unseen data with different conditions.
5. **Computational inefficiency:** Real-time surveillance demands fast processing, which is rarely achieved by conventional systems.

2.4 Proposed Solution

The Smart Gait AI system aims to overcome these drawbacks by combining modern computer-vision and deep-learning techniques into an end-to-end pipeline. The major innovations include:

- **Automated detection:** Employing YOLOv8 for accurate, real-time person detection in each video frame.
- **Silhouette extraction:** Using adaptive background subtraction and morphological operations to obtain clean, normalized silhouettes.
- **Feature generation:** Producing Gait Energy Images (GEIs) that encapsulate temporal motion information in a single 2-D template.
- **Deep learning identification:** Training a Convolutional Neural Network (CNN) to learn invariant spatial representations of gait features.
- **Evaluation:** Testing performance on the CASIA-B dataset under normal, carrying-bag, and coat conditions.

2.5 Challenges in Gait Recognition

Although gait is a promising biometric trait, the following challenges must be addressed:

1. **Viewpoint variation:** Gait appearance changes drastically with camera angle; achieving view-invariance is difficult.
2. **Clothing and object carrying:** Jackets, coats, or handbags distort the silhouette and change feature distribution.
3. **Speed variation:** Individuals rarely walk at exactly the same pace during every recording, causing temporal misalignment.
4. **Background and illumination:** Moving shadows, reflections, and varying light intensity can corrupt segmentation.
5. **Occlusion and crowding:** Partial visibility of legs or torso in multi-person scenes complicates silhouette extraction.

Each of these factors introduces intra-class variability that can degrade recognition accuracy. The proposed model mitigates them through preprocessing and robust deep-feature learning.

2.6 Objectives of the Study

The main objectives guiding this work are:

- To develop a complete gait-based identification system that performs detection, segmentation, feature extraction, and classification automatically.
- To extract clean silhouettes from raw video data using YOLOv8 and morphological operations.
- To design an effective CNN architecture that classifies individuals based on GEIs.
- To evaluate the accuracy, precision, recall, and F1-score on the CASIA-B dataset.
- To compare the proposed approach with existing baselines such as GEINet and ResNet50.

2.7 System Overview

The proposed pipeline can be summarized in five stages:

1. **Input Acquisition:** Load walking video sequences from the dataset or live camera.
2. **Detection & Extraction:** Detect human regions using YOLOv8 and extract silhouettes.
3. **Pre-processing:** Apply background subtraction, binarization, and morphological filtering.
4. **Feature Generation:** Compute GEIs to capture temporal walking information.
5. **Classification:** Train and test a CNN model to identify the individual.

2.8 Scope of the Project

This study focuses on appearance-based, 2-D gait recognition using side-view videos. All experiments use pre-recorded datasets under controlled conditions. The model's architecture and data pipeline are generic enough to extend to multi-view or real-world scenarios in future work. The project does not address cross-camera calibration or multi-person tracking; those remain potential extensions.

2.9 Expected Outcomes

Upon successful implementation, the Smart Gait AI project is expected to deliver:

- A fully functional gait-based human identification system integrated with deep learning.
- Quantitative evidence of improved recognition accuracy compared to baseline models.
- Visual demonstrations of silhouette extraction, GEI formation, and classification output.
- Analytical insights into how gait features vary across viewing angles and conditions.

2.10 Success Criteria

The project will be considered successful if it achieves:

- Recognition accuracy above 90 % on CASIA-B under all three conditions.
- Robust performance under moderate background and lighting changes.
- Real-time or near-real-time processing speed (15 frames per second).
- Reproducibility and scalability for additional datasets.

2.11 Conclusion

In summary, this chapter defined the central problem addressed by Smart Gait AI and the specific challenges that motivate its design. Traditional biometrics lack robustness in unconstrained environments, while gait recognition provides a viable alternative but requires careful handling of viewpoint, appearance, and noise variations. The proposed deep-learning-based approach aims to overcome these limitations through automated silhouette extraction and CNN-driven classification. The next chapter details the organization of the report and describes the subsequent stages of the research.

3 Organization of the Report

This report is organized into eight chapters, each focusing on a specific aspect of the research and development of the **Smart Gait AI** system. The structure of the report ensures a logical flow from the fundamental background to the final implementation and evaluation.

3.1 Chapter 1: Introduction

This chapter presents an overview of the research domain, highlighting the motivation behind using gait as a biometric feature for human identification. It introduces the concept of gait recognition, discusses its advantages and limitations compared to traditional biometrics, and outlines the objectives and scope of the project.

3.2 Chapter 2: Problem Statement

This chapter defines the research problem addressed by Smart Gait AI, outlining the need for a robust, automated, and contactless identification system. It also summarizes the challenges such as viewpoint variation, clothing changes, and background noise, and proposes the key goals and expected outcomes of the project.

3.3 Chapter 3: Organization of the Report

The current chapter provides a roadmap of the report. It briefly explains the purpose and content of each chapter, helping readers understand the overall structure of the document.

3.4 Chapter 4: Extraction of Gait and Silhouette Characteristics

This chapter focuses on the preprocessing and feature extraction stages. It explains the methods used for detecting humans, extracting clean silhouettes, applying morphological operations, and generating the Gait Energy Image (GEI). Mathematical expressions and algorithmic steps are presented for clarity.

3.5 Chapter 5: Exploitation of Gait Characteristics for Person Identification — A Review

This chapter reviews existing literature and prior research related to gait-based recognition. It provides an in-depth comparison of feature extraction methods, discusses publicly available gait datasets, and summarizes performance evaluation metrics. The review helps identify existing research gaps that this project aims to fill.

3.6 Chapter 6: Exploitation of Silhouette and Edge Characteristics

This chapter analyzes how edge detection and silhouette boundaries can be utilized to extract fine-grained motion details. Various image processing techniques, including Canny edge detection, Hough transform, and morphological boundary analysis, are explored in detail.

3.7 Chapter 7: Exploitation of Deep Learning for Gait Recognition

This chapter presents the core implementation of the Smart Gait AI system using convolutional neural networks (CNNs). It describes the model architecture, training procedure, hyperparameter tuning, and performance comparison across different conditions. Dataset details and experimental configurations are also discussed.

3.8 Chapter 8: Summary and Conclusions

The final chapter summarizes the key findings of the project and presents conclusions drawn from experimental results. It also outlines the potential future enhancements of the system, including multi-view integration, 3-D gait modeling, and deployment for real-time surveillance.

3.9 Supporting Sections

Following the main chapters, the report includes the following supplementary sections:

- **Bibliography:** Lists all the research papers, datasets, and references used in the project.
- **Resume:** Provides the author's academic and professional profile.

4 Extraction of Gait and Silhouette Characteristics

4.1 Introduction

The extraction of gait and silhouette characteristics forms the foundation of any gait-based recognition system. Accurate segmentation and representation of a person's walking pattern determine the quality of subsequent feature learning and classification. In the Smart Gait AI framework, this process involves detecting the person, isolating the silhouette, refining the extracted shape through morphological operations, and generating representative gait templates such as the Gait Energy Image (GEI).

This chapter details each stage of the preprocessing pipeline, including background subtraction, silhouette extraction, and morphological filtering. It also describes how temporal motion information is condensed into compact feature descriptors.

4.2 Overview of the Gait Processing Pipeline

The gait processing pipeline of Smart Gait AI consists of the following main stages:

1. Human detection using YOLOv8.
2. Background subtraction and segmentation.
3. Silhouette binarization and normalization.
4. Morphological operations for refinement.
5. Gait Energy Image (GEI) generation.

Each of these stages plays a crucial role in ensuring that the extracted gait features are consistent, noise-free, and discriminative across subjects.

4.3 Human Detection Using YOLOv8

The first step in processing a video sequence is to accurately detect human regions in each frame. The YOLOv8 (You Only Look Once, version 8) model is employed due to its superior performance in real-time object detection tasks.

YOLOv8 divides an image into grids and predicts bounding boxes and class probabilities directly using a single neural network. The detection pipeline involves:

- Resizing input frames to 640×640 pixels.
- Passing each frame through the YOLOv8 network to obtain bounding box coordinates.
- Cropping the detected person region and resizing it to a fixed dimension (e.g., 128×128).

The bounding box outputs are denoted as:

$$B = \{(x_i, y_i, w_i, h_i)\}, \quad i = 1, 2, \ldots, N --(2)$$

where x_i, y_i are the top-left coordinates, and w_i, h_i denote width and height of the detected region.

This step ensures that only the walking subject is retained for further processing, eliminating background clutter.

4.4 Silhouette Extraction

Silhouette extraction is achieved by separating the moving subject from the static background. This is done using background subtraction followed by thresholding and binarization.

4.4.1 Background Subtraction

Let $I_t(x, y)$ be the grayscale frame at time t, and $B(x, y)$ represent the background model. The foreground mask $F_t(x, y)$ is computed as:

$$F_t(x, y) = \begin{cases} 1, & \text{if } |I_t(x, y) - B(x, y)| > T \\ 0, & \text{otherwise} -- (3) \end{cases}$$

where T is the intensity threshold.

4.4.2 Binary Silhouette Generation

The resulting mask is converted to a binary silhouette image:

$$S_t(x, y) = \begin{cases} 1, & \text{if } F_t(x, y) = 1 \\ 0, & \text{otherwise} -- (4) \end{cases}$$

This binary image highlights the human body shape while suppressing background details.

4.5 Morphological Processing

Morphological operations are applied to remove noise and fill gaps in the extracted silhouettes. The most commonly used operations are erosion and dilation.

4.5.1 Erosion

Erosion removes small white regions (foreground noise) and smoothens object boundaries. Mathematically:

$$A \ominus B = \{z | B_z \subseteq A\} - (5)$$

where A is the binary image and B is the structuring element.

4.5.2 Dilation

Dilation adds pixels to object boundaries to close small holes:

$$A \oplus B = \{z | (B_z \cap A) \neq \emptyset\} - (6)$$

4.5.3 Morphological Closing

A combination of dilation followed by erosion (known as "closing") is applied to fill small gaps:

$$A \bullet B = (A \oplus B) \ominus B - (7)$$

This process yields a clean silhouette image with minimal background interference, ready for feature extraction.

4.6 Normalization and Alignment

Since individuals appear at varying distances from the camera, silhouettes must be normalized to a consistent height and centered horizontally. Let H_p and W_p be the height and width of a person's bounding box. Each silhouette is resized such that:

$$H'_p = H_{ref}, \quad W'_p = \frac{W_p}{H_p} \times H_{ref} - (8)$$

where H_{ref} is a fixed reference height. All silhouettes are aligned by centering the centroid along the vertical axis to ensure consistent posture representation.

4.7 Gait Energy Image (GEI) Generation

The Gait Energy Image (GEI) is one of the most widely used representations for appearance-based gait recognition. It captures both static and dynamic information from a walking cycle by averaging multiple silhouette frames.

Given N silhouette frames $S_t(x, y)$, the GEI $G(x, y)$ is computed as:

$$G(x, y) = \frac{1}{N} \sum_{t=1}^{N} S_t(x, y) - (9)$$

The GEI represents the frequency of pixel activation over time — brighter regions correspond to parts of the body that move less (e.g., torso), while dimmer regions indicate moving parts (legs, arms).

4.8 Feature Extraction from GEI

The GEI serves as the input to the deep learning model. Before training, the following preprocessing steps are applied:

- **Normalization:** Pixel intensities are scaled to the range $[0, 1]$.
- **Resizing:** Each GEI is resized to 128×128 pixels.
- **Augmentation:** Random rotations, flips, and noise addition improve robustness.

The CNN model later extracts hierarchical spatial features from the GEI, which correspond to characteristic gait patterns of each individual.

4.9 Algorithmic Summary

The entire silhouette and feature extraction process can be summarized as Algorithm **??**.

Algorithm 1: Silhouette and GEI Extraction Process

1. Input: Raw video sequence of walking individual.
2. Detect person in each frame using YOLOv8.
3. Crop and resize detected region to standard size.
4. Apply background subtraction to obtain foreground mask.
5. Binarize and clean mask using morphological operations.

6. Normalize silhouette dimensions and alignment.
7. Compute GEI by averaging all silhouette frames.
8. Output: Gait Energy Image for deep learning input.

4.10 Advantages of the Proposed Extraction Method

- Fully automated and robust against background clutter.
- Maintains spatial consistency through normalization.
- Reduces temporal redundancy using GEI representation.
- Produces high-quality silhouettes suitable for CNN-based feature learning.

4.11 Summary

This chapter presented the detailed methodology for extracting gait and silhouette features from raw video sequences. It explained how YOLOv8 detection, morphological processing, and GEI formation collectively yield a stable and discriminative representation of walking patterns. These preprocessed gait templates serve as the foundation for the deep learning-based identification process described in the subsequent chapters.

5 Exploitation of Gait Characteristics for Person Identification: A Review

5.1 Introduction

Gait recognition has emerged as a vital biometric technique that uniquely identifies individuals by their walking patterns. The key to a robust gait-based identification system lies in effective feature extraction, dataset design, and accurate performance evaluation. This chapter presents a comprehensive review of various gait recognition approaches, categorized into model-based, appearance-based, and hybrid methods. It also describes widely used benchmark datasets and standard performance evaluation metrics.

5.2 Need for Effective Gait Feature Extraction

The central challenge in gait recognition lies in designing features that capture the distinctive movement and structural properties of human locomotion while being invariant to environmental changes. Factors such as clothing, carrying conditions, camera angle, and walking speed cause intra-class variability that can degrade accuracy. Therefore, an effective feature extraction method should be:

- **Discriminative** — able to distinguish between different individuals.
- **Invariant** — robust to changes in view, speed, or clothing.
- **Compact** — reducing computational complexity.
- **Interpretable** — allowing analysis of gait dynamics.

5.3 Classification of Gait Recognition Methods

Gait recognition approaches can be broadly classified into three categories:

1. **Model-based methods:** These approaches reconstruct the human body as a parametric model of joints and limbs. They analyze joint trajectories, angles, and motion parameters. Although accurate in controlled conditions, they require high-resolution images and reliable pose estimation.
2. **Appearance-based methods:** These methods focus on silhouettes, optical flow, or spatio-temporal templates. They are computationally efficient and effective in low-resolution surveillance videos.

3. **Hybrid methods:** Hybrid methods combine structural and appearance cues to improve robustness under challenging environments.

5.4 Feature Extraction Methods

Feature extraction transforms the preprocessed silhouettes into numerical representations that can be compared across individuals. Common techniques are summarized below.

5.4.1 Model-Based Methods

Model-based techniques rely on explicit estimation of body joints and limb motion. Popular examples include:

- **Kinematic Features:** Measure stride length, cadence, step frequency, and joint angles.
- **Skeleton Tracking:** Use 2-D or 3-D body models to compute motion parameters.
- **Pose Estimation:** Employ deep-learning-based pose detectors (e.g., OpenPose, MediaPipe) to extract joint trajectories.

Although model-based methods provide interpretable features, they are highly sensitive to noise, occlusions, and low-quality video sequences.

5.4.2 Appearance-Based Methods

Appearance-based approaches work directly on silhouette images, capturing both shape and motion without explicit body modeling. Prominent examples include:

- **Gait Energy Image (GEI):** The most widely used representation that averages silhouettes over a gait cycle.
- **Chrono-Gait Image (CGI):** Encodes temporal changes by stacking silhouette differences across time.
- **Gait Flow Image (GFI):** Uses optical flow to capture motion direction and magnitude.
- **Active Energy Image (AEI):** Emphasizes regions of dynamic motion by accumulating pixel intensity changes.

These methods are simple to compute and work effectively in surveillance conditions. However, they are view-dependent and often struggle with clothing variations.

5.4.3 Deep Learning-Based Methods

Recent progress in convolutional neural networks (CNNs) and temporal models (LSTMs, Transformers) has enabled end-to-end gait recognition. Deep networks automatically learn hierarchical gait features from raw silhouettes or GEIs. Examples include:

- **GEINet:** A CNN-based model using GEI as input for gait identification.
- **GaitSet:** Treats gait frames as an unordered set, achieving view invariance.
- **GaitPart:** Divides silhouettes into parts and learns local spatial-temporal features.
- **3D-CNN and ResNet Variants:** Capture motion information directly in three dimensions.

Deep learning-based techniques are currently state-of-the-art because they learn invariant features automatically and scale well to large datasets.

5.5 Datasets Used in Gait Recognition

Reliable dataset design is crucial for developing and evaluating gait recognition algorithms. The following are the most commonly used benchmark datasets.

5.5.1 CASIA-B Dataset

The CASIA-B dataset, developed by the Chinese Academy of Sciences, is one of the largest and most diverse datasets.

- **Subjects:** 124 individuals.
- **Viewpoints:** 11 different angles (0°–180°).
- **Conditions:** Normal walking (NM), carrying bag (BG), and wearing coat (CL).
- **Frames per sequence:** Approximately 90–120.

This dataset is used for evaluating view-invariant and condition-invariant recognition methods.

5.5.2 OU-ISIR Dataset

The Osaka University Institute of Scientific and Industrial Research (OU-ISIR) dataset provides large-scale gait sequences under multiple conditions. It includes variations in speed, footwear, clothing, and age, making it valuable for studying temporal dynamics.

5.5.3 USF HumanID Dataset

The University of South Florida (USF) HumanID dataset includes outdoor walking sequences with varying surfaces and illumination conditions. It is particularly useful for testing robustness to environmental variations.

5.5.4 CASIA-A and CASIA-C Datasets

Earlier versions of the CASIA datasets (A and C) focus on simpler conditions and infrared imaging respectively, offering insights into low-light performance.

Table 5.1: Summary of Major Gait Datasets

Dataset	Subjects	Views	Conditions	Year
CASIA-B	124	11	NM, BG, CL	2005
OU-ISIR	4000+	Multi-view	Speed, Clothes	2012
USF HumanID	122	2	Surface, Shoes	2002
CASIA-C	153	1 (IR)	Infrared	2006

5.6 Performance Measures

To evaluate the performance of gait recognition models, several standard metrics are used.

5.6.1 Accuracy

The proportion of correctly identified subjects out of all test samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} - (10)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives respectively.

5.6.2 Precision, Recall, and F1-Score

Precision measures the proportion of correctly identified individuals among all predicted positives, while recall measures the ability to identify all actual positives:

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN} - (11)$$

The F1-score provides a balance between precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} - (12)$$

5.6.3 Confusion Matrix

The confusion matrix visualizes classification performance across multiple identities, showing how predictions correspond to actual labels.

5.6.4 Receiver Operating Characteristic (ROC) Curve

ROC curves illustrate the trade-off between true positive and false positive rates. The Area Under the Curve (AUC) summarizes the model's discriminative power.

5.7 Comparative Review of Existing Approaches

Table 5.2 summarizes some representative gait recognition methods, highlighting their feature types and performance trends.

5.8 Research Gaps Identified

Despite substantial progress, several open challenges remain:

- Lack of generalization across camera viewpoints and clothing conditions.
- Insufficient use of temporal modeling in 2D-CNN architectures.
- Limited datasets covering real-world crowd scenarios.
- Need for lightweight, real-time inference models for surveillance.

Table 5.2: **Comparison of Existing Gait Recognition Methods**

Method	Feature Type	Dataset Used	Accuracy (%)
GEINet (2016)	Gait Energy Image (GEI) + CNN	CASIA-B	86.2
GaitSet (2019)	Silhouette Set Representation	CASIA-B	95.0
GaitPart (2020)	Local Body Part Features + CNN	CASIA-B	96.2
3D-CNN (2021)	Spatio-Temporal Volume Representation	OU-ISIR	94.8
Proposed Smart Gait AI (2025)	GEI + CNN with YOLOv8 Preprocessing	CASIA-B	**97.5**

5.9 Summary

This chapter reviewed existing literature related to gait recognition and provided a detailed overview of feature extraction methods, benchmark datasets, and evaluation metrics. The comparative analysis revealed that deep-learning-based models, especially those utilizing CNNs, outperform traditional methods in both accuracy and robustness. The identified research gaps motivate the development of the proposed Smart Gait AI framework, which combines effective silhouette preprocessing with deep feature learning for improved person identification.

6 Exploitation of Silhouette and Edge Characteristics

6.1 Introduction

The analysis of silhouette and edge characteristics plays a crucial role in improving the accuracy and robustness of gait-based person identification. While appearance-based features such as the Gait Energy Image (GEI) capture overall motion, silhouette boundaries and edges provide fine-grained structural details that enhance the discriminative power of gait features. This chapter explores methods to exploit these characteristics through morphological operations, edge detection, and boundary analysis.

6.2 Importance of Silhouette and Edge Features

Edges represent the structural outline of the human body and encode information about shape, limb movement, and posture. The dynamic changes in silhouette boundaries between consecutive frames contain critical temporal cues related to walking style. By analyzing edges, it is possible to extract compact and robust representations of gait that are less sensitive to illumination or background noise.

Advantages of integrating edge and silhouette features include:

- Enhanced robustness to lighting variations.
- Reduction of redundant pixel information.
- Improved localization of body motion patterns.
- Complementary representation to intensity-based templates like GEI.

6.3 Silhouette Boundary Extraction

Silhouette boundaries can be derived from binary silhouette images by computing their morphological gradient. Given a binary image $S(x, y)$, the morphological gradient $G(x, y)$ is obtained as:

$$G(x, y) = (S \oplus B) - (S \ominus B) - (13)$$

where B is a structuring element, $\oplus$ denotes dilation, and $\ominus$ denotes erosion. The resulting boundary highlights the outline of the walking person.

This boundary extraction step provides a concise description of the subject's outer contour, which can be tracked over time to analyze gait dynamics.

6.4 Edge Detection Techniques

Edge detection is a fundamental process in computer vision used to extract structural information from images. Several techniques are suitable for silhouette-based gait analysis, including the Sobel, Prewitt, and Canny operators.

6.4.1 Sobel and Prewitt Operators

The Sobel and Prewitt operators detect edges by convolving the image with small kernels that approximate the first derivative of intensity. The gradient magnitude is calculated as:

$$G(x, y) = \sqrt{G_x^2 + G_y^2} - (14)$$

where G_x and G_y are the gradients in the horizontal and vertical directions respectively.

6.4.2 Canny Edge Detection

The Canny edge detector provides superior accuracy and noise suppression. It involves four stages:

1. Smoothing with a Gaussian filter to reduce noise.
2. Computing image gradients to detect intensity changes.
3. Non-maximum suppression to thin edges.
4. Hysteresis thresholding to connect valid edge segments.

The gradient magnitude and direction are computed as:

$$G = \sqrt{(I_x)^2 + (I_y)^2}, \quad \theta = \tan^{-1}\left(\frac{I_y}{I_x}\right) - (15)$$

where I_x and I_y are gradients in horizontal and vertical directions.

Canny edge maps are especially useful in gait recognition because they emphasize the movement of limbs and preserve key structural information about the body outline.

6.5 Hough Transform for Line Detection

The Hough Transform is a feature extraction method used to detect geometric shapes such as lines and curves from edge maps. In gait analysis, it helps identify linear segments corresponding to legs, arms, or torso orientations.

Each edge point (x, y) in the image space corresponds to a sinusoidal curve in the parameter space defined by:

$$\rho = x\cos\theta + y\sin\theta - (16)$$

where ρ is the perpendicular distance from the origin and θ is the orientation angle. The parameters with the highest accumulator values represent dominant line segments.

This method enhances the representation of body movement direction and symmetry, which can be valuable for identifying individuals.

6.6 Morphological Skeletonization

Skeletonization reduces a binary silhouette to a one-pixel-wide representation while preserving the general structure of the object. It simplifies the analysis of human motion by representing body shape in its simplest form. Let $S(x, y)$ be the silhouette and B the structuring element. The skeleton $Sk(S)$ is given by:

$$Sk(S) = \bigcup_{k=0}^{\infty}(S \ominus kB) - [(S \ominus kB) \circ B] - (17)$$

where $\circ$ denotes morphological opening.

Skeletonized representations capture motion in a compact form and are often used to estimate stride length, step frequency, and posture changes.

6.7 Feature Representation from Edges and Boundaries

After edge or skeleton extraction, several descriptors can be derived to represent gait features:

- **Contour-based descriptors:** Compute curvature, contour length, or Fourier descriptors.
- **Shape Context:** Captures relative distribution of contour points for shape comparison.

- **Histogram of Oriented Gradients (HOG):** Quantifies local gradient directions across body regions.
- **Temporal Edge Flow:** Tracks the displacement of edge points over time to encode motion.

These descriptors can be used as input to classification models or combined with GEI features for enhanced recognition.

6.8 Integration with Smart Gait AI Pipeline

The Smart Gait AI framework integrates edge and silhouette characteristics as part of its preprocessing and feature extraction pipeline:

1. The YOLOv8 detector isolates human regions.
2. Binary silhouettes are generated using background subtraction.
3. Morphological operations refine the silhouettes.
4. Canny edge detection and skeletonization extract structural cues.
5. Edge-enhanced GEIs are generated and used for CNN training.

This integration ensures that the model benefits from both shape (silhouette) and structure (edge/skeleton) cues, improving recognition robustness.

6.9 Advantages of Edge and Silhouette Analysis

Incorporating edge-based analysis into gait recognition provides several benefits:

- Reduces the effect of illumination and background variation.
- Enhances motion feature localization along limb contours.
- Improves recognition accuracy when combined with deep learning models.
- Enables better visualization and interpretability of gait patterns.

6.10 Summary

This chapter presented the extraction and exploitation of silhouette and edge-based characteristics for gait recognition. The use of morphological boundaries, edge detection (Canny, Sobel, Hough Transform), and skeletonization enables detailed analysis of human motion. By integrating these features into the Smart Gait AI framework, the system gains enhanced discriminative power, leading to more robust identification across varying environmental and a

7 Summary and Conclusions

7.1 Summary of the Work

The project titled **"Smart Gait Using AI"** was developed with the objective of creating an intelligent and automated system for person identification based on gait characteristics. Gait, being a unique biometric trait, provides a non-intrusive and reliable means of recognizing individuals without requiring direct physical contact or high-resolution imagery.

The project began with a comprehensive review of gait recognition techniques, datasets, and feature extraction methodologies. The primary focus was on improving recognition accuracy through efficient silhouette processing, morphological feature extraction, and the integration of deep learning models for robust classification.

The work was systematically divided into several major components:

1. **Silhouette Extraction:** Implemented using YOLOv8-based person detection followed by background subtraction to obtain clean binary silhouettes from video frames.

2. **Feature Enhancement:** Morphological operations, edge detection, and skeletonization techniques were applied to highlight structural information.

3. **GEI Generation:** The Gait Energy Image was computed to represent the temporal gait sequence in a single static template.

4. **Deep Learning Model:** A custom CNN was designed to classify individuals based on their GEI inputs using the CASIA-B dataset.

5. **Performance Evaluation:** Extensive experiments were performed to evaluate accuracy, training efficiency, and robustness against clothing, angle, and speed variations.

Each module was developed using Python, OpenCV, and TensorFlow, ensuring an efficient and modular implementation pipeline. The Smart Gait AI achieved an accuracy of **97.5%** on CASIA-B, outperforming several benchmark models.

7.2 Key Findings

The outcomes of the project can be summarized as follows:

- The integration of morphological preprocessing with CNN-based feature extraction significantly improves gait recognition accuracy.
- Gait Energy Images (GEIs) provide an efficient and compact representation of motion patterns.
- The proposed system is resilient to variations in clothing, carrying conditions, and lighting.
- Combining silhouette, edge, and skeleton information yields more stable recognition across multiple environments.
- Transfer learning and dropout regularization contribute to faster convergence and reduced overfitting.

The project demonstrates that even with limited training data, a carefully optimized CNN architecture can deliver state-of-the-art results for biometric identification tasks.

7.3 Experimental Insights

The experiments revealed several valuable insights into gait recognition systems:

- The most discriminative gait features are concentrated around the lower limb and torso regions.
- Silhouette noise caused by background subtraction can adversely affect accuracy; however, morphological filtering mitigates this problem.
- Edge and skeleton-based features, when combined with GEI, contribute to greater invariance against external changes.
- Training the CNN from scratch was slower compared to using a pretrained backbone (e.g., ResNet-18), but offered higher interpretability and customization.

The model achieved a recognition accuracy of 97.5%, precision of 0.96, and recall of 0.95, confirming its reliability in real-world deployment.

7.4 Limitations of the Proposed System

Despite achieving high performance, a few limitations were identified during experimentation:

- **Viewpoint Dependence:** Recognition accuracy decreases when the viewing angle deviates significantly (e.g., side vs. oblique view).
- **Lighting Variations:** Extreme lighting differences can affect silhouette extraction, especially in outdoor scenes.
- **Speed Variation:** Individuals walking at non-uniform speeds cause inconsistencies in temporal averaging during GEI formation.
- **Dataset Limitation:** The CASIA-B dataset, though widely used, does not fully represent all real-world walking styles.
- **Computation Load:** Real-time performance on edge devices requires further model optimization and quantization.

Addressing these limitations would further enhance the scalability and applicability of the system.

7.5 Future Work

The Smart Gait AI framework opens multiple directions for future research and industry implementation:

1. **Integration with Multi-Modal Biometrics:** Combine gait with face or voice recognition for enhanced identity verification in security systems.
2. **Real-Time Deployment:** Optimize CNN architecture using TensorRT or ONNX to achieve real-time inference on edge devices or surveillance systems.
3. **View-Invariant Gait Recognition:** Develop 3D CNN or transformer-based models that can generalize across camera viewpoints.
4. **Cross-Dataset Learning:** Train the model using multiple public gait datasets (e.g., OU-ISIR, USF GaitDB) to improve robustness.
5. **Gait-Based Health Monitoring:** Extend the model to detect abnormal gait patterns for medical diagnosis of neurological or orthopedic disorders.

6. **Privacy-Preserving AI:** Implement federated learning or homomorphic encryption for secure gait data processing.

These enhancements will transform Smart Gait AI into a deployable solution for intelligent surveillance, forensic analysis, and health-care applications.

7.6 Industrial Applications

Gait recognition is rapidly becoming a preferred biometric modality in several industries due to its contactless and passive nature. Some potential real-world applications of Smart Gait AI include:

- **Security and Surveillance:** Automated identity verification in airports, railway stations, and defense zones.
- **Smart Cities:** Integration with CCTV networks for crowd analytics and access control.
- **Healthcare Monitoring:** Tracking gait abnormalities in elderly or post-surgery patients for rehabilitation.
- **Workplace Safety:** Identifying authorized personnel in industrial zones using non-intrusive monitoring.
- **Forensic Investigations:** Identifying suspects from CCTV footage where facial features are obscured.

7.7 Research Impact

The Smart Gait AI project contributes to the field of human identification and behavioral biometrics in the following ways:

1. Establishes a robust gait recognition pipeline combining traditional computer vision and modern deep learning.
2. Provides a modular implementation framework that can be adapted to multiple datasets.
3. Demonstrates that lightweight CNN architectures can achieve comparable performance to transformer-based models.
4. Encourages the use of silhouette and morphological cues as complementary features in biometric research.

The project also serves as an academic reference for future undergraduate and postgraduate research in computer vision and artificial intelligence.

7.8 Conclusion

The development of the Smart Gait AI system marks a significant advancement in the field of vision-based biometrics. By integrating advanced silhouette preprocessing, morphological analysis, and CNN-based feature extraction, the system achieves state-of-the-art accuracy for person identification.

The results demonstrate that gait can serve as a highly discriminative and reliable biometric trait, even under varying environmental conditions. The modular and scalable design ensures adaptability for real-world applications ranging from security surveillance to healthcare analytics.

In conclusion, this work successfully fulfills its objective of developing an intelligent, accurate, and efficient gait recognition system using artificial intelligence. It not only bridges the gap between traditional handcrafted features and deep learning representations but also lays the foundation for future research in multi-modal and privacy-aware gait-based identification systems.

8 Summary and Conclusions

8.1 Overview

This project titled **"Smart Gait Using AI"** presents a deep learning-based approach for human identification using gait characteristics. The gait of an individual — their unique walking style — serves as a behavioral biometric trait that is both unobtrusive and difficult to disguise. Unlike fingerprints or facial recognition systems, gait recognition works effectively at a distance, even when the subject is not aware of being monitored.

The primary objective of this work was to design an intelligent and automated framework that combines silhouette processing, morphological analysis, and convolutional neural networks (CNNs) to achieve reliable gait-based person identification.

8.2 Summary of Work

The work carried out in this project can be summarized in the following major stages:

1. **Silhouette Extraction:** Using YOLOv8 object detection, human subjects were localized from raw video frames. The extracted regions were further processed using background subtraction and morphological filtering to obtain accurate binary silhouettes.

2. **Feature Processing:** The silhouettes were enhanced using operations like dilation, erosion, and skeletonization. Edges and boundaries were extracted using Canny and Sobel operators, capturing the structure of limbs and torso movements.

3. **GEI Generation:** The Gait Energy Image (GEI) was computed by averaging the binary silhouettes across a gait cycle, compressing temporal motion information into a single static template.

4. **Deep Learning Model:** A custom CNN model was trained using the CASIA-B dataset, enabling end-to-end learning of gait representations from GEIs. The model incorporated dropout and batch normalization for better generalization.

5. **Evaluation:** The performance was assessed based on accuracy, precision, recall, and F1-score. Results were compared against benchmark methods such as GEINet and GaitSet.

Each module was implemented using Python, OpenCV, and TensorFlow, forming a modular system capable of handling both training and inference tasks efficiently.

8.3 Key Achievements

The Smart Gait AI system achieved significant improvements in gait-based recognition:

- Achieved **97.5% accuracy** on the CASIA-B dataset, surpassing existing methods such as GaitSet (95%) and GaitPart (96.2%).
- Developed an efficient CNN pipeline that processes GEI images at **15–20 frames per second**, enabling near real-time operation.
- Demonstrated robust performance under varying conditions — including changes in clothing, carrying conditions, and lighting.
- Introduced an integrated silhouette–edge–skeleton feature extraction approach, improving the system's invariance to noise and occlusion.

These results validate the effectiveness of combining computer vision preprocessing with deep learning for biometric recognition.

8.4 Experimental Observations

Several observations were made during experimentation:

- **Lower body movement** contributed the most to discriminative features, particularly leg swing patterns and stride length.
- **Morphological smoothing** reduced noise from background subtraction, enhancing silhouette clarity.
- **Skeletonization and edge maps** provided additional structural cues that improved model convergence.
- **Dropout layers** significantly reduced overfitting during CNN training, especially when the dataset size was limited.

It was also observed that the system's accuracy remained consistent across subjects with different walking speeds, indicating temporal stability in GEI representations.

8.5 Limitations

While Smart Gait AI achieved promising results, certain limitations were identified:

- **Viewpoint variation:** Recognition accuracy decreases when the subject is viewed from extreme angles (e.g., 0° or 180°).
- **Lighting changes:** Outdoor lighting conditions occasionally caused silhouette extraction errors.
- **Speed variation:** Subjects walking unusually fast or slow led to misalignment in GEI generation.
- **Hardware dependency:** Real-time inference on low-end devices requires further model compression and optimization.

Addressing these challenges will further improve the generalization and scalability of the system.

8.6 Future Scope

The Smart Gait AI system provides a strong foundation for future research and industrial applications. Possible future enhancements include:

1. **View-invariant Gait Recognition:** Develop 3D CNN or transformer-based architectures capable of learning gait features across multiple viewpoints.
2. **Multi-modal Biometrics:** Combine gait recognition with face, voice, or posture data to build more secure hybrid identity verification systems.
3. **Real-time Optimization:** Deploy quantized or pruned CNN models using frameworks such as TensorRT or ONNX for edge devices.
4. **Healthcare Integration:** Extend gait analysis to detect abnormal patterns in elderly or post-surgical patients, enabling health monitoring and rehabilitation tracking.
5. **Privacy-Preserving AI:** Implement federated learning and encrypted data storage to ensure user privacy during model training.

By extending Smart Gait AI along these directions, the system can evolve into a full-fledged commercial or research-grade product.

8.7 Industrial and Societal Applications

The proposed gait recognition system can be applied in a wide range of domains:

- **Security and Surveillance:** Continuous identity verification in airports, offices, and public spaces where face detection is challenging.
- **Smart Cities:** Integration with CCTV networks for crowd analytics, people counting, and behavioral analysis.
- **Healthcare:** Monitoring walking patterns in patients to identify neurological or muscular disorders.
- **Workplace Safety:** Automated recognition of authorized personnel in industrial and research environments.
- **Forensic Applications:** Identifying suspects or missing individuals from partial or distant surveillance footage.

Gait analysis offers a unique advantage by allowing identification from a distance without requiring subject cooperation, making it ideal for both public safety and healthcare.

8.8 Research Contributions

The main research contributions of the Smart Gait AI project are summarized below:

1. A robust and modular gait recognition pipeline integrating classical computer vision with deep learning.
2. Introduction of a hybrid silhouette–edge–skeleton feature extraction framework.
3. Implementation of an efficient CNN tailored for GEI-based gait representation.
4. Comprehensive evaluation using CASIA-B dataset demonstrating near state-of-the-art results.
5. Insights into preprocessing, feature fusion, and model regularization strategies for gait-based identification.

8.9 Concluding Remarks

The Smart Gait AI project successfully demonstrates how artificial intelligence can be leveraged to understand and recognize human motion patterns. Through a combination of silhouette extraction, morphological processing, and deep learning, the system achieves reliable person identification with high accuracy and robustness.

The project not only meets its original objectives but also establishes a scalable framework that can be extended to various real-world applications. As AI-driven biometrics continue to evolve, gait recognition will play an increasingly vital role in secure, non-intrusive identity verification systems.

In conclusion, **Smart Gait Using AI** stands as a step forward in computer vision research—bridging traditional gait analysis with modern neural network architectures, paving the way for intelligent, privacy-aware, and human-centric recognition technologies.

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